

AthenaK Figure-3 Brill Evolution: Telegrapher and Gamma-driver Damping Diagnostics

Authenticated numerical investigation bundle

14 August 2026

Abstract

This report consolidates five matched AthenaK evolutions of the $A = -0.047$ Brill initial data used in the Figure-3 reproduction study. Four runs compare prospectively selected scale-invariant telegrapher damping fields. A fifth, single controlled diagnostic changes only the Gamma-driver shift damping from $\eta = 2 \max |K|$ to fixed $\eta = 2$. The fixed- η control strongly implicates the global $\max |K|$ shift scaling in the baseline's late stiffness, but the control itself develops a different central curvature/constraint pathology and is neither a robust solution nor scale invariant. No coefficient sweep, threshold relaxation, floor, clipping, or selective rerun was used.

1 Common Figure-3 configuration

All cases use the same IrisK global 48×32 coefficient payload, directly interpolated by AthenaK without an M0 intermediate-domain reconstruction. The Brill amplitude is $A = -0.047$ and the imported ADM mass is 2.660301967997158. The initial lapse is pre-collapsed, $\alpha = \psi^{-2}$. Evolution uses the $\rho \geq 0$ Cartoon half-plane, N_{128} , sixth-order spatial operators, RK4, CFL 0.15, KO dissipation 0.02, nine available AMR levels, and $d_{\chi_{\max}} = 0.02$. The strict-positive chi gate is unchanged and `floor_chi=false`. The lapse is the telegrapher gauge with $\tau = \kappa = 1$ and the domain-max $|K|$ prescription unless stated otherwise. Z4c damping remains enabled with its existing max- K scaling.

The source for all five runs is:

```
commit: 2a8ad80e02279769a99fe279b7a33516bc6c8d0d
tree: 67709c405a1169a15643cb933eec5353cd216243
Kokkos: 6739bc623081648af9e752b616d9671527922cbf
```

2 Local telegrapher-scale comparison

The four prospectively fixed inverse-length fields were

$$\mu \in \left\{ \max_{\Omega} |K|, |K(x)|, \sqrt{K_{ij}K^{ij}}, \sqrt{\gamma^{ij}\partial_i\chi\partial_j\chi} \right\}.$$

With $K_{\star} = \max_{\Omega} |K|$, the conceptual scaling is

$$Q = \frac{\mu}{K_{\star}}, \quad \tau_{\text{eff}} = \frac{\tau}{K_{\star}}, \quad \kappa_{\text{eff}} = \frac{\kappa}{K_{\star}}.$$

The kernel evaluates the cancelled ratios $Q/\tau_{\text{eff}} = \mu/\tau$ and $\kappa_{\text{eff}}/\tau_{\text{eff}} = \kappa/\tau$, providing a finite extension through the time-symmetric initial slice.

telegrapher damping field	terminal t/M	terminal τ/M	rejected parents
$\max_{\Omega} K $	10.162940	6.201069	6412
$ K(x) $	8.907275	5.472376	13184
$\sqrt{K_{ij}K^{ij}}$	9.193945	5.751011	12
$ \nabla\chi _{\gamma}$	8.358252	5.016554	7320

All four runs stopped at the unchanged positive-chi boundary-prolongation gate. The rejected-parent count is an inventory at different physical states, not a stability ranking. The domain-maximum case remains the best comparator among these four; none is a qualified Figure-3 evolution.

3 Fixed-shift control

The controlled follow-up changes exactly one runtime setting:

$$\text{z4c/shift_eta_max_K=true} \longrightarrow \text{z4c/shift_eta_max_K=false}.$$

Thus the baseline has $\eta_{\text{shift}} = 2 \max |K|$ while the control has fixed $\eta_{\text{shift}} = 2$. The source, executable, input, initial-data payload, telegrapher lapse, Z4c damping, mesh, AMR, timestep controls, and numerical thresholds are identical.

Define the explicit RK damping stiffness number

$$S_{\beta} = \Delta t \eta_{\text{shift}}.$$

The baseline reaches $S_{\beta} = 5.55889$, exceeding the approximate RK4 negative-real-axis limit 2.8, immediately before its chi-gate failure. The fixed- η control never exceeds $S_{\beta} = 0.075$.

At the baseline terminal time, the fixed- η evolution remains controlled: its combined constraint norm is 136.6 rather than 2.40×10^{11} , $\max |K| = 3.49$ rather than 1.90×10^4 , sampled $\max |\tilde{\Gamma}| = 0.237$ rather than 1288, and it uses 32 rather than 290 MeshBlocks. Central lapse and central proper time remain closely matched.

The fixed- η case avoids the chi rejection and reaches $t = 16.73892M$, $\tau = 10.37631M$: $6.57598M$ or 64.7% farther in coordinate time. It then fails a different strict diagnostic after cycle 2333 because axis-central physical support becomes nonfinite or invalid. The last finite row already has $\max |K| = 1.0689 \times 10^4$, axis Kretschmann 7.6154×10^{15} , and combined constraint norm 8.3439×10^7 . The frozen telemetry does not identify whether lapse, constraint density, or curvature reconstruction becomes invalid first.

4 Interpretation and qualification boundary

The controlled result is strong evidence that global $\max |K|$ -scaled Gamma-driver damping is a major contributor to the baseline stiffness and $t \simeq 10.16M$ failure. The causal case is not based merely on later survival: the sole parameter change prevents the RK4 damping violation and suppresses the accompanying $\tilde{\Gamma}$, constraint, and AMR explosion at matched time.

It does not establish that shift scaling is the only failure mechanism. Fixed $\eta = 2$ introduces a dimensionful scale, violates the target scale-invariance principle, and ends in a central curvature/constraint pathology. Consequently the result supports redesigning the shift damping scale; it does not support adopting fixed η , weakening the chi gate, or claiming Figure-3 reproduction. The report makes no convergence, horizon, or production qualification claim.

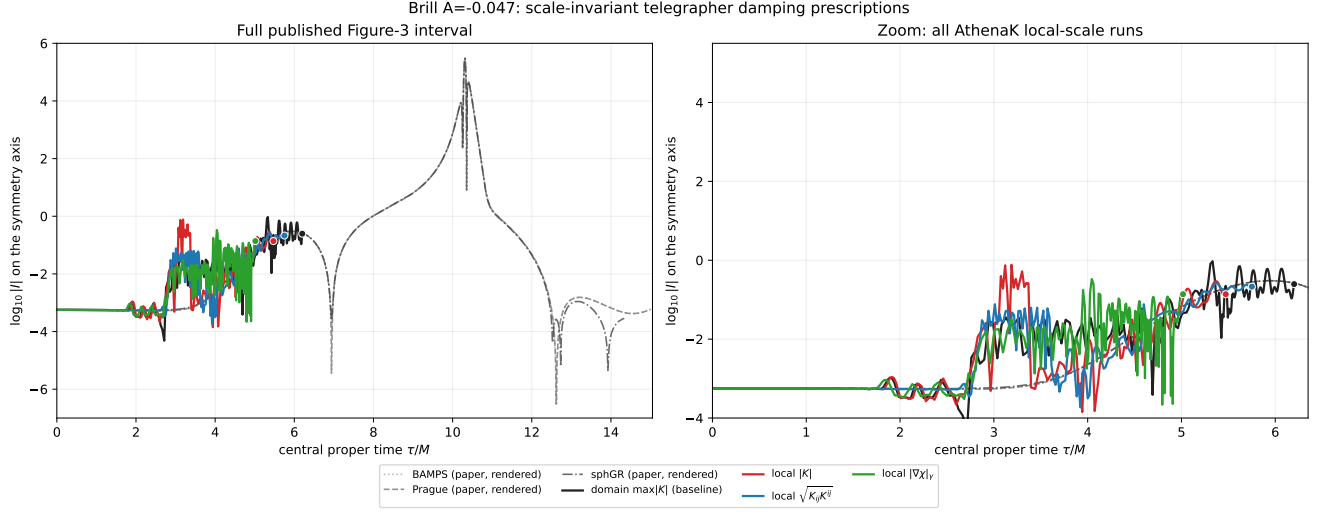


Figure 1: The four scale-invariant telegrapher damping choices plotted on the rendered Figure-3 axes. All terminate before covering the published interval.

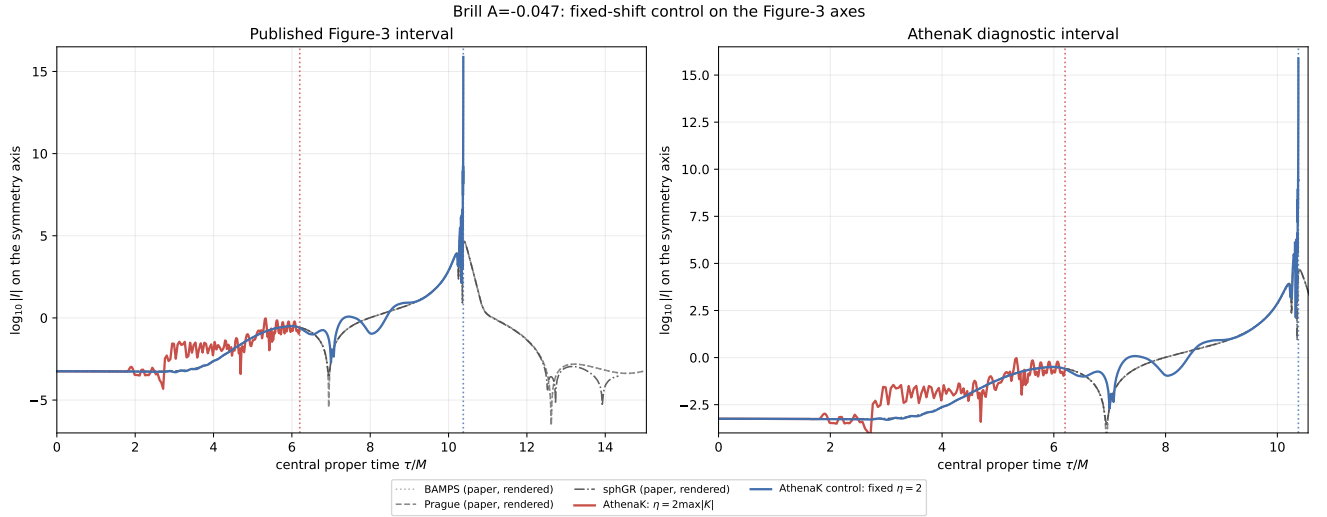


Figure 2: The fixed- η control added directly to the Figure-3 reproduction axes. The dotted vertical lines show the last finite baseline and control history rows. Longer survival is diagnostic evidence, not reproduction or convergence.

Figure-3 Brill diagnostic: shift stiffness and gauge fields

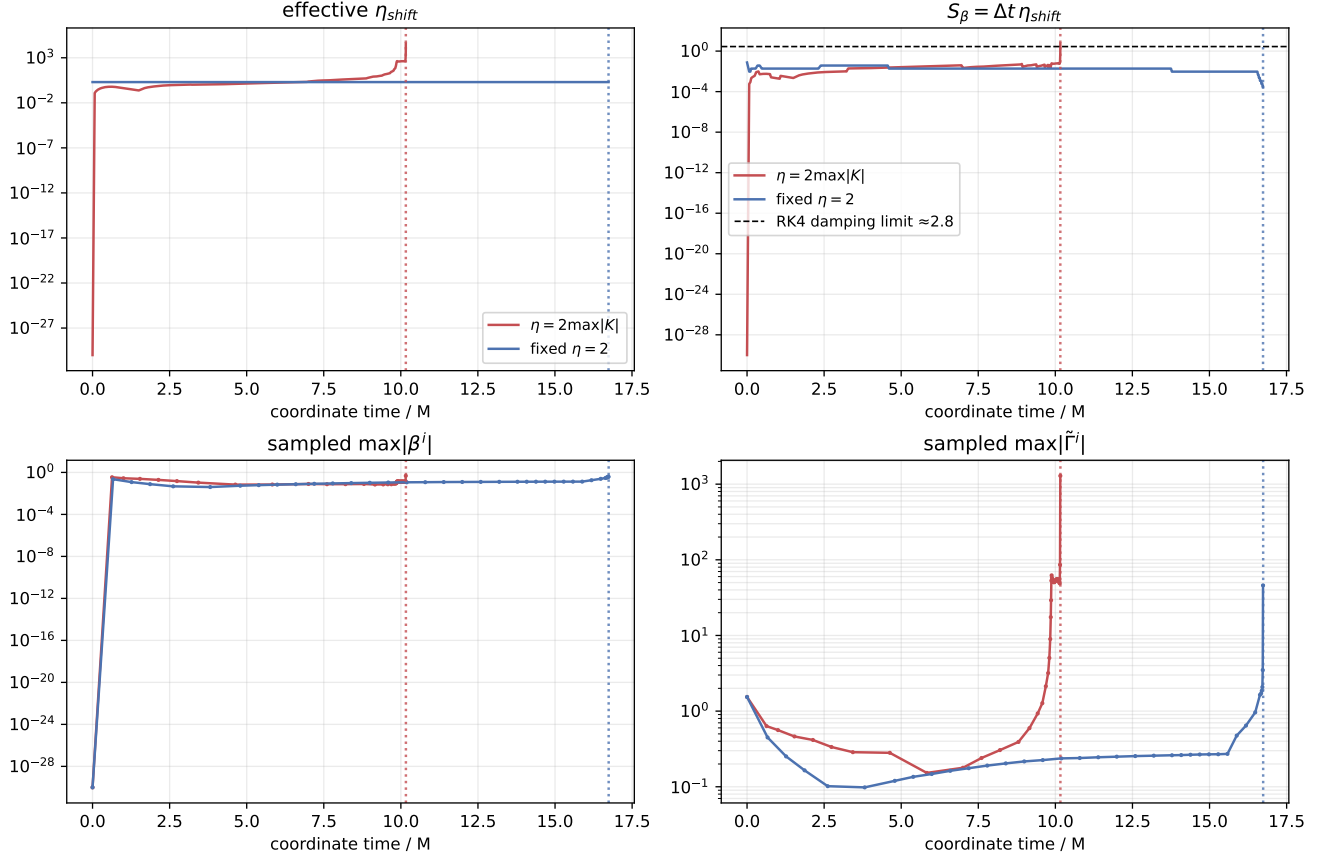


Figure 3: Effective shift damping, stiffness number, and sampled shift and conformal-connection extrema. The fixed control removes the baseline RK4 damping-axis violation but later develops its own growth.

Figure-3 Brill diagnostic: constraint norms

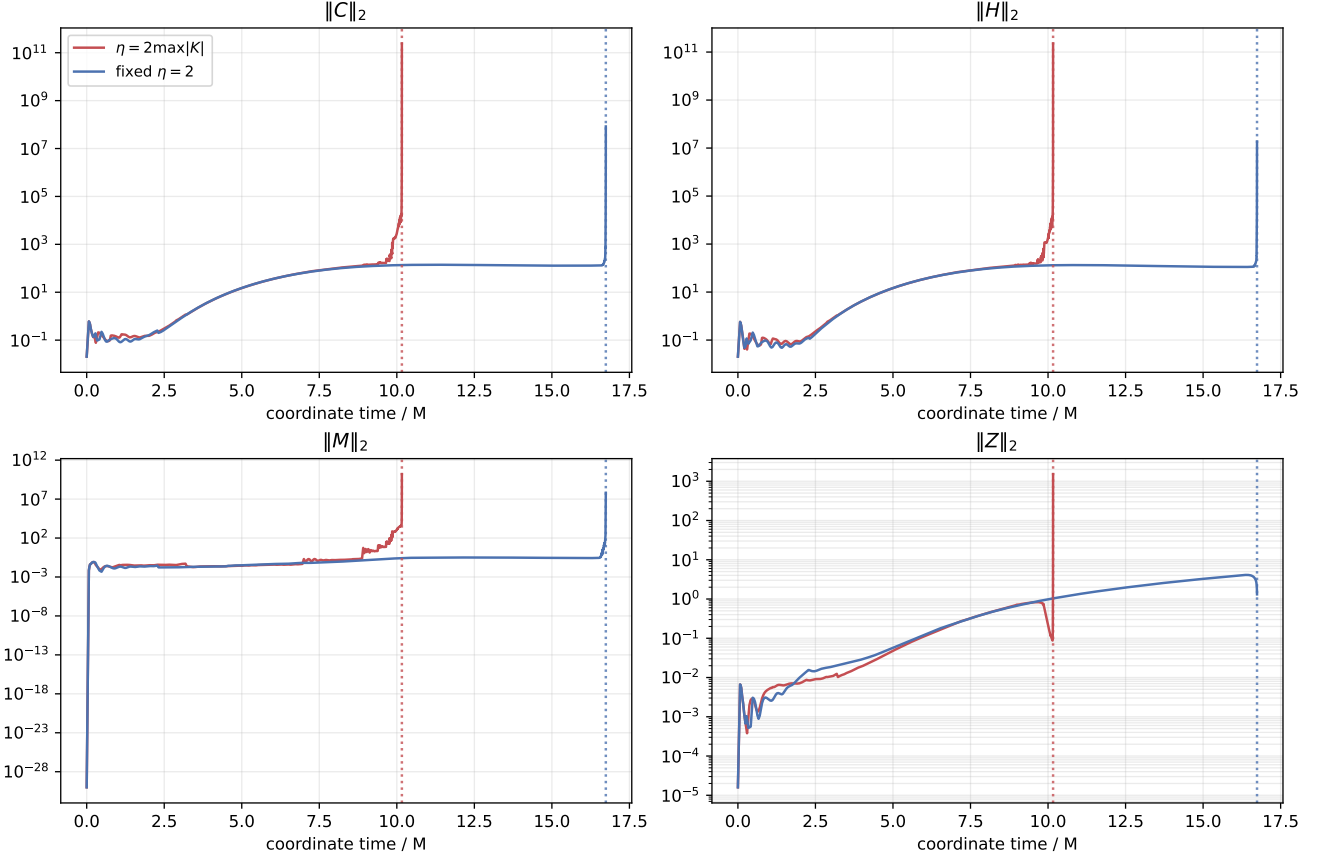


Figure 4: Constraint histories for the matched baseline and fixed- η control. The control stays orders of magnitude smaller at the baseline failure time, then grows rapidly near its distinct terminal event.

Figure-3 Brill diagnostic: collapse trajectory and AMR

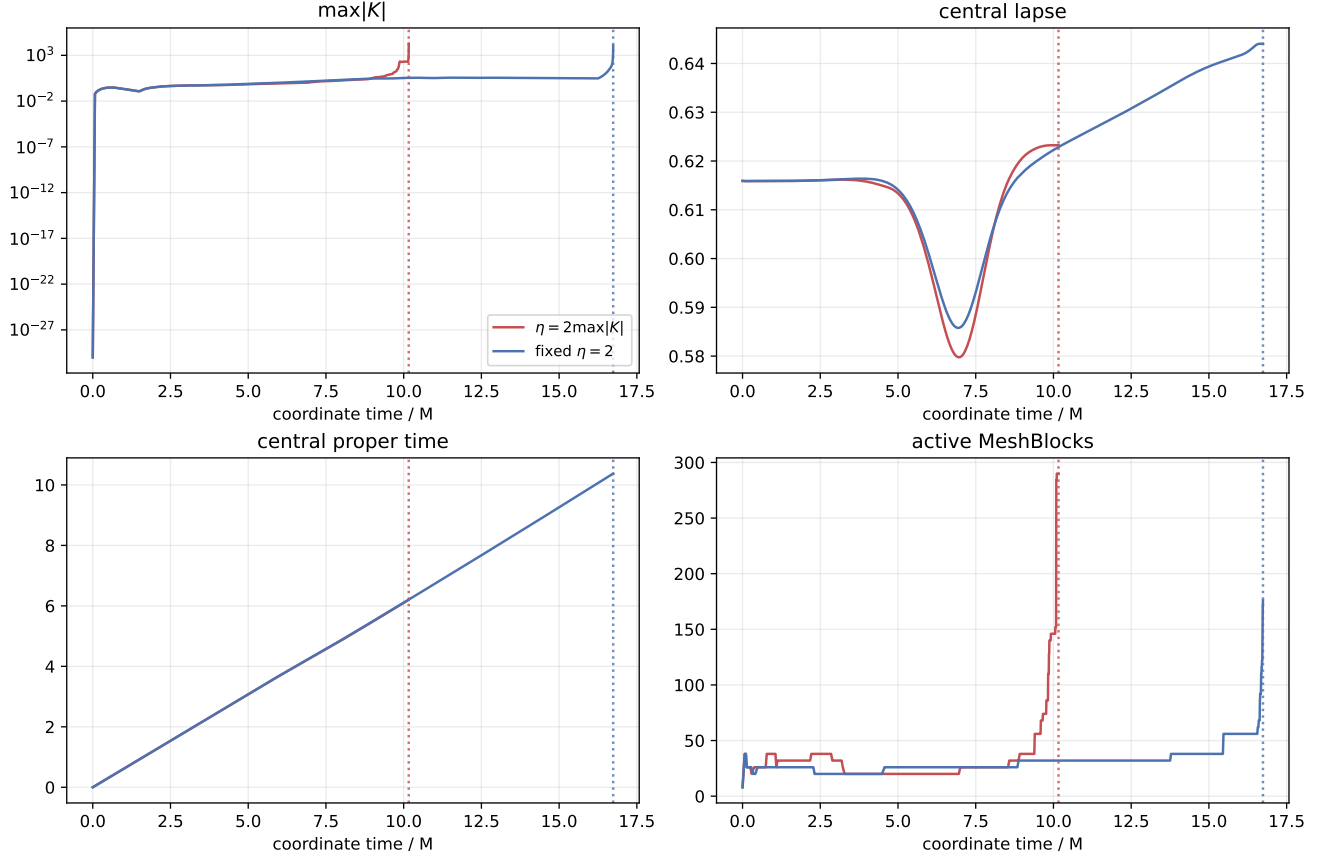


Figure 5: Collapse trajectory and AMR inventory. The two evolutions track closely at early times, separate before the baseline failure, and both end in unqualified late-time pathologies.

The smallest useful next reasoning task is mathematical rather than another parameter sweep: construct a smooth positive inverse-length functional for the shift whose induced $\Delta t \eta$ remains bounded while preserving mass rescaling, and identify a prospective diagnostic that distinguishes gauge stiffness from the later central curvature failure.

5 Provenance

The four-mode campaign is Perlmutter job 56955603. The fixed- η control is job 56966125. Its key SHA-256 identities are:

```
executable: 8325bd78f4851f04e6810aaf9fb5ecfc4007d876c3a3011a296512ffbd7875be
root manifest: 29acd8986ce53afbb8bf472b70e09f554e224c640977528a421ffb0558713fba
detached manifest: ce2136a1a81f1cd06cab592240147359732805aea72b837f8417c6ad7fb8c07d
```

The fixed-control result JSON and reduced CSVs in this directory are sufficient to reproduce every plot in this report without native AthenaK binaries.